

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

CHEMISTRY

Paper 1 Multiple Choice

8872/01

Wednesday 29 August 2013
50 minutes

Additional Materials: Multiple Choice Answer Booklet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

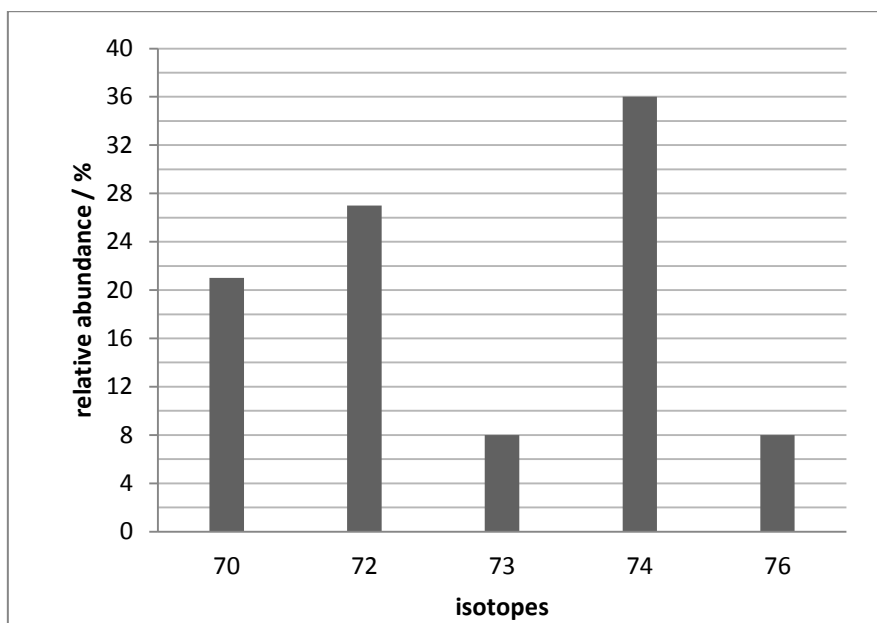
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Section A

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

- 1** Element **X** has five isotopes. The mass spectrum of element **X** is shown below.

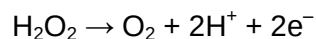


What is the relative atomic mass, rounded to the nearest 1 decimal place, of element **X**?

- A** 72.7 **B** 73.8 **C** 74.2 **D** 75.1
- 2** Hydrogen peroxide, H_2O_2 , is used in some cleaning products and bleaching agents.

When 20.0 cm^3 of hydrogen peroxide was titrated with $0.200 \text{ mol dm}^{-3}$ KMnO_4 , it requires 12.0 cm^3 of KMnO_4 for the first pink colour to appear.

Given the following half-equation,



what is the concentration of hydrogen peroxide used ?

- A** $0.100 \text{ mol dm}^{-3}$ **B** $0.200 \text{ mol dm}^{-3}$
C $0.300 \text{ mol dm}^{-3}$ **D** $0.400 \text{ mol dm}^{-3}$

- 3** An antiknocking agent is a gasoline additive used to reduce engine knocking and increase the efficiency of combustion of fuel when ignition occurs. A lead additive, PbC_xH_y , was added to petrol as an antiknocking agent. 0.00310 mol of PbC_xH_y requires 1.04 dm³ of oxygen for complete combustion and 594 cm³ of carbon dioxide gas and lead(IV) oxide are produced. All volumes were measured at r.t.p.

What is the empirical formula of the lead additive?

- A** PbC_8H_6 **B** $\text{PbC}_8\text{H}_{10}$ **C** $\text{PbC}_8\text{H}_{20}$ **D** $\text{PbC}_8\text{H}_{24}$

- 4** Species containing one or more unpaired electrons and can be attracted by an external magnetic field are said to be paramagnetic.

Which of the following species is **not** paramagnetic?

- A** Cr^{3+} **B** K^+ **C** Ni^{2+} **D** V^{3+}

- 5** What do the ions $^{31}\text{P}^{3-}$ and $^{32}\text{S}^{2-}$ have in common?

- A** Both ions have 34 electrons.
B Both ions have the same number of protons.
C Both ions have the more protons than neutrons.
D $^{32}\text{S}^{2-}$ has the same number of protons as the number of neutrons in $^{31}\text{P}^{3-}$.

- 6** Elemental oxygen exists in the atmosphere in two forms, O_2 and O_3 . Even though O_2 and O_3 are both made up of oxygen atoms, they have different physical and chemical properties.

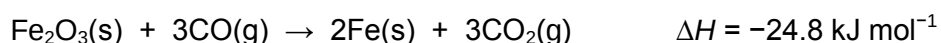
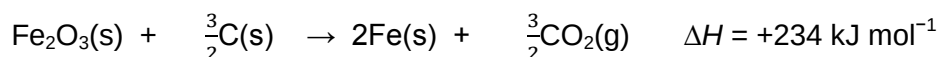
Which of the following best explains why O_3 is more soluble in water as compared to O_2 ?

- A** O_3 is more stable and thus it is able to react more readily with water as compared to O_2 .
B O_3 is able to form more extensive van der Waals' forces with water molecules as compared to O_2 .
C O_3 is able to form permanent dipole-permanent dipole interaction with water while O_2 is not able to.
D O_3 has more lone electrons as compared to O_2 and therefore forms stronger ion-dipole interaction with water.

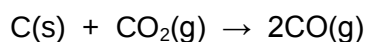
- 7 Beryllium difluoride reacts readily with trimethylamine, $(\text{CH}_3)_3\text{N}$ to form a stable addition product. However, nitrogen trifluoride has no reaction with trimethylamine.

Which of the statements is **not** true?

- A The bond angle in the addition product is 109.5° .
 - B The molar ratio for the reaction between beryllium difluoride and $(\text{CH}_3)_3\text{N}$ is 1:1.
 - C Nitrogen trifluoride does not react as the nitrogen atom lacks energetically accessible orbitals for reaction.
 - D The bonding formed between beryllium difluoride and trimethylamine is dative covalent bond.
- 8 In a blast furnace, carbon or carbon monoxide can be used to reduce iron (III) oxide.



What is the enthalpy change of reaction in kJ mol^{-1} for the following reaction?

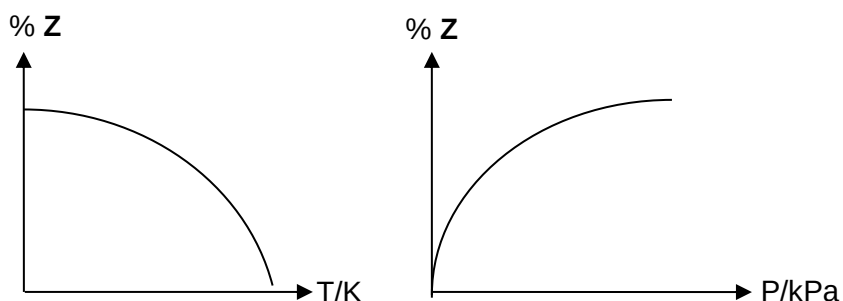


- | | |
|----------|----------|
| A +139.5 | B +172.5 |
| C +388.2 | D +443.2 |
- 9 Which of the following is **not** an endothermic process?
- A sublimation of a solid
 - B dissociation of a diatomic molecule into atoms
 - C photosynthesis
 - D human respiration

- 10 When X is reacted with Y, the following equilibrium is established.



The two graphs below show how the percentage of Z varies with temperature and pressure at equilibrium.



Which of the following responses concerning the reaction is correct?

- | | ΔH | No. of moles gases |
|---|------------|--------------------|
| A | negative | reactant > product |
| B | negative | product > reactant |
| C | positive | reactant > product |
| D | positive | product > reactant |

- 11 The values of ionic product of water, K_w , at two different temperatures are given below.

temperature/ °C	$K_w/\text{mol}^2\text{dm}^{-6}$
25	1.00×10^{-14}
100	5.50×10^{-13}

Which of the following statement is correct for water at 100 °C?

- A $[\text{H}^+] > [\text{OH}^-]$
 B $[\text{OH}^-] < [\text{H}^+]$
 C $\text{pH} > 7$
 D $\text{pH} < 7$

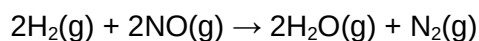
- 12** Consider the following equilibrium in the human body:



Which species present in the human body removes excess $\text{H}^+(\text{aq})$ to help keep the blood pH constant at about 7.4?

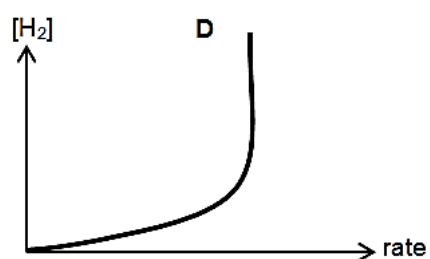
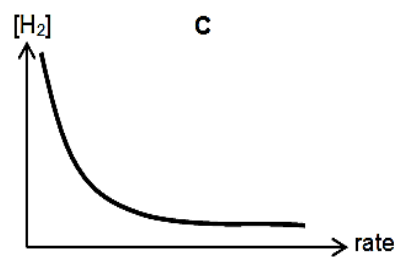
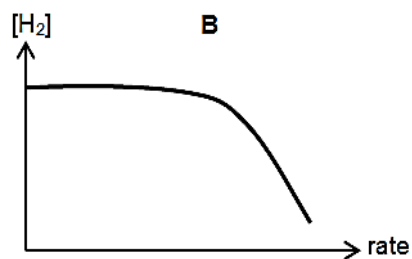
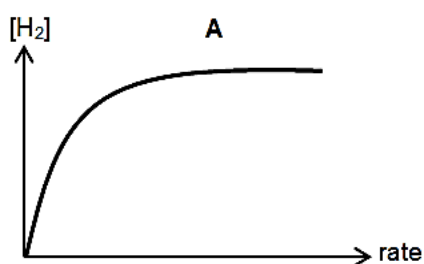
- A** CO_2 **B** H_2O **C** H_2CO_3 **D** HCO_3^-

- 13** Consider the following reaction

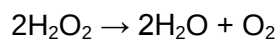


In an experiment, the reaction rate is found to be proportional to $[\text{H}_2]$ at low concentrations of hydrogen but independent of $[\text{H}_2]$ when the concentration is high.

Which graph represents the experimental results?



- 14** The decomposition of hydrogen peroxide, a first order reaction, gives water and oxygen as products.

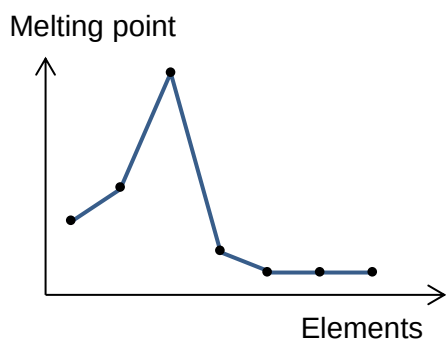


In an experiment, 0.20 mol of pure H_2O_2 was put into an evacuated flask and it was found that there was 0.050 mol of H_2O_2 left 30 minutes later.

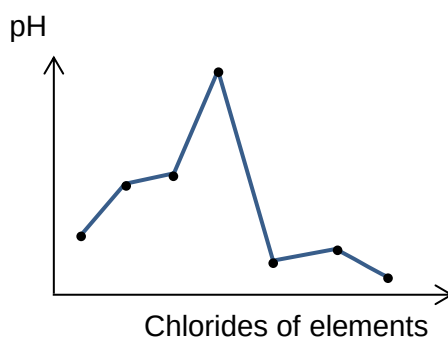
Which statement is true?

- A** It took 15 minutes for the amount of H_2O to rise from 0 to 0.10 mol.
- B** The amount of H_2O in the flask went up by 2 times in the first 30 minutes.
- C** At 30 minutes, the amount of H_2O formed is the same as the amount of H_2O_2 left.
- D** After 60 minutes, the amount of H_2O formed is 4 times more than the amount of H_2O_2 left.
- 15** Element **Y** can be found in period three of the periodic table. The oxide of **Y** dissolves in water to give an alkaline solution while the chloride of **Y** dissolves in water to give an acidic solution. What is the identity of element **Y**?
- | | |
|--------------------|---------------------|
| A Sodium | B Magnesium |
| C Aluminium | D Phosphorus |

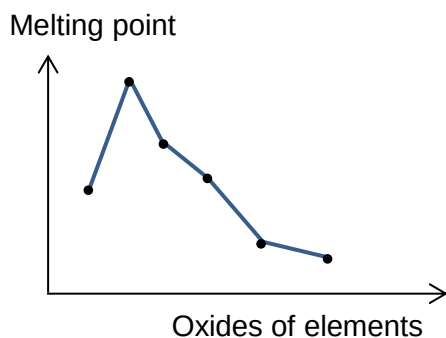
- 16 Which of the following show the correct trend across the elements in period 3 of the periodic table?



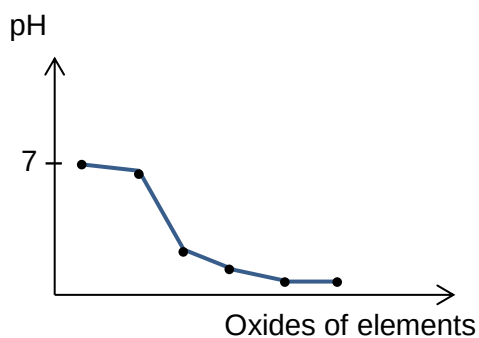
A



B



C



D

- 17 How many possible alcohols are there in a compound with a molecular formula $C_4H_{10}O$?

A 4

B 5

C 6

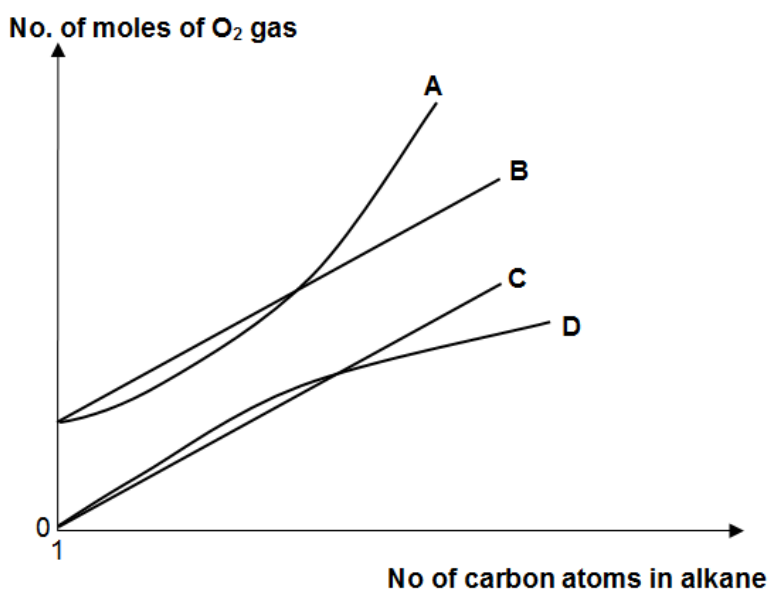
D 7

- 18 When one but-1-ene, $CH_3CH_2CHCH_2$, molecule undergoes electrophilic addition with one $Br_2(g)$, how many bonds are broken and made in the reaction?

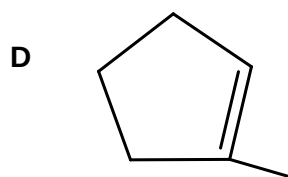
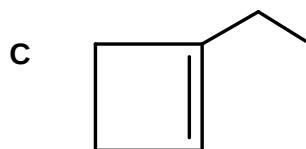
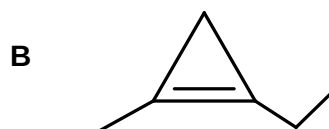
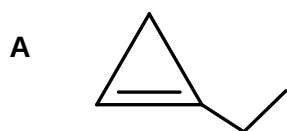
	No. of σ bonds broken	No. of new σ bonds made	No. of π bonds broken	No. of new π bonds made
A	1	2	1	0
B	0	3	2	0
C	0	2	1	2
D	1	3	2	2

- 19 Alkanes undergo complete combustion to form carbon dioxide and water at room temperature and pressure.

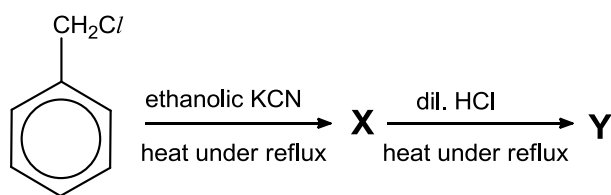
Which of the following graphs shows the relationship between the number of carbon atoms in an alkane and the number of moles of oxygen gas needed for complete combustion of one mole of the alkane?



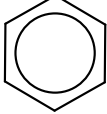
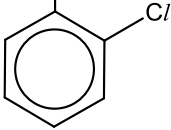
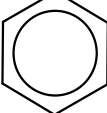
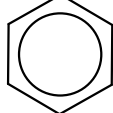
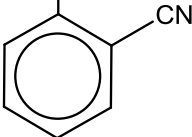
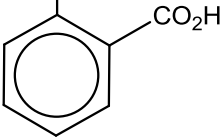
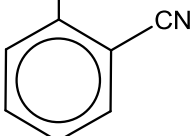
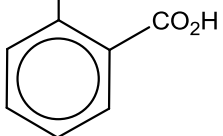
- 20 Hydrocarbon **X**, produces $\text{HO}_2\text{CCH}_2\text{CH}_2\text{COCH}_2\text{CH}_3$ upon heating with an excess of hot concentrated acidified $\text{KMnO}_4(\text{aq})$. What is a possible identity of **X**?



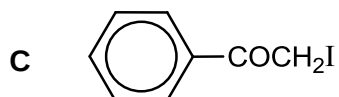
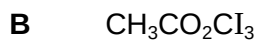
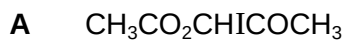
21 A synthesis reaction was carried out as follows:



Which organic products are formed from this synthesis?

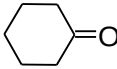
	X	Y
A	CH_2CN 	CH_2CN 
B	CH_2CN 	$\text{CH}_2\text{CO}_2\text{H}$ 
C	CH_2Cl 	CO_2H 
D	CH_2Cl 	CH_2Cl 

22 Which compounds will not produce CHI_3 on warming with alkaline $\text{I}_2(\text{aq})$?



- 23 Compounds **P**, **Q** and **R** all react with 2,4-dinitrophenylhydrazine but only one of them will cause a reduction in the oxidation number of the metal present in Fehling's reagent.

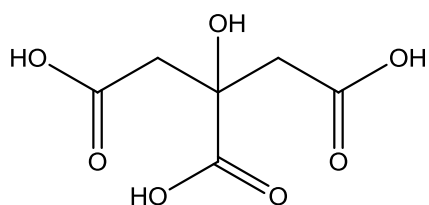
What are the possible identities of **P**, **Q** and **R**?

	P	Q	R
A	$\text{C}_6\text{H}_5\text{CHO}$	CH_3CHO	CH_3COCH_3
B	CH_3CONH_2	$\text{C}_6\text{H}_5\text{CHO}$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$
C	CH_3COCH_3	$\text{C}_6\text{H}_5\text{CHO}$	
D	$\text{CH}_3\text{CH}_2\text{CHO}$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_3$	CH_3COCH_3

- 24 Which of the following is arranged in order of decreasing pK_a ?

- A** $\text{ClCH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CH}_2\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{OH}$
- B** $\text{CH}_3\text{CH}_2\text{OH}$, $\text{BrCH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CH}_2\text{CO}_2\text{H}$, $\text{ClCH}_2\text{CO}_2\text{H}$
- C** $\text{CH}_3\text{CH}_2\text{OH}$, $\text{BrCH}_2\text{CH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CO}_2\text{H}$, $\text{ClCH}_2\text{CO}_2\text{H}$
- D** $\text{BrCH}_2\text{CO}_2\text{H}$, $\text{ClCH}_2\text{CO}_2\text{H}$, $\text{BrCH}_2\text{CH}_2\text{CO}_2\text{H}$, $\text{CH}_3\text{CH}_2\text{OH}$

- 25 Oranges contain citric acid with the following structural formula.



Which of the following reagents react completely with 1 mol of citric acid?

- A** 3 mol of Na_2CO_3
- B** 3 mol of SOCl_2
- C** 4 mol of Na
- D** 4 mol of NaOH

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Sodium hypobromite, NaOBr, is commonly used as germicide and antiparasites.

What type of bonding is present in NaOBr?

- 1** Ionic bond
- 2** Covalent bond
- 3** Van der Waals forces

27 The rate equation for the reaction $\text{W(aq)} + 3\text{X(aq)} \rightarrow \text{Y(aq)}$ is

$$\text{Rate} = k [\text{H}^+][\text{X}]^2$$

Which of the following is **true**?

- 1** H^+ is a catalyst in the reaction.
- 2** When $[\text{W}]$ is halved and $[\text{X}]$ is doubled, the reaction rate increases by 4 times.
- 3** The rate constant has the unit of $\text{mol}^{-1} \text{dm}^3 \text{s}^{-1}$.

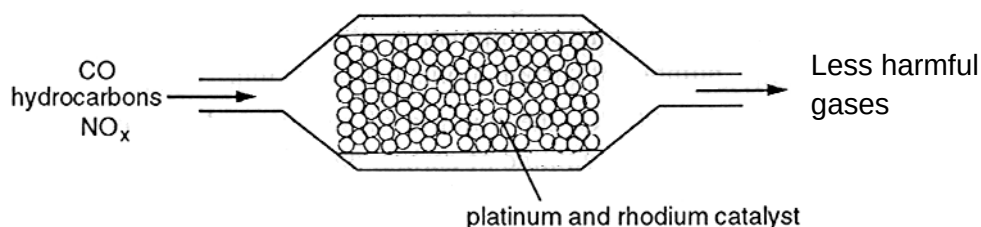
28 Arsenic trioxide is an inorganic compound with the formula As_2O_3 . It is the main precursor to other arsenic compounds, including organoarsenic compounds.

As_2O_3 appears as a white solid at room temperature and has similar properties to P_4O_6 .

Which of the following statements regarding arsenic trioxide is true?

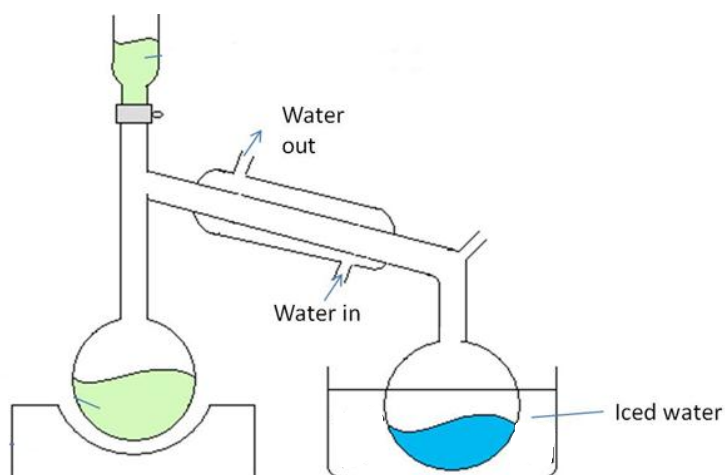
- 1** It is hydrolysed by water to give an acidic solution.
- 2** It dissolves in alkaline solution.
- 3** It has a relatively low boiling point.

- 29** The diagram below shows a section of a catalytic converter in the exhaust system of a car. Harmful pollutant gases are converted into less harmful gases before they are emitted into the atmosphere.



Which processes take place in the catalytic converter?

- 1** Combustion of CO to form CO₂.
 - 2** CO and NO reacting together to form N₂ and CO₂.
 - 3** CO and hydrocarbons reacting together to form CO₂ and H₂O
- 30** The diagram below illustrates an experimental set-up for preparation of organic compounds.



Which of the following can be prepared with the use of the above set-up?

- 1** CH₃CH₂CHO from CH₃CH₂CH₂OH with acidified K₂Cr₂O₇
- 2** CH₃CH₂Cl from CH₃CH₂OH with PCl₅
- 3** CH₃CH₂CO₂H from CH₃CH₂CH₂OH with acidified KMnO₄

~ END OF PAPER ~